A APPENDIX

Algorithm 1 PLPP

function BATCH-UPDATING($I, Prompts$)
    # Get The Features of the current batch, labels and output distribution of prompts.
    $I_f \leftarrow image_encoder(I)$
    $T_f \leftarrow text_encoder(Prompts)$
    $\tilde{T}_f \leftarrow text_encoder_before_proj(Prompts)$
    $prompts_label \leftarrow generate_label(prompts)$
    $output \leftarrow lm_head(\tilde{T}_f)$

    # Calculate the losses.
    $L_{CE} \leftarrow$ CE$(I_f @ T_f.transpose(), label)$
    $L_{PPL} \leftarrow$TEXT-PPL-LOSS$(output, prompts_label)$
    $L_{PLPP} \leftarrow L_{CE} + \lambda L_{PPL}$

    # Update the prompts.
    $Prompts \leftarrow$BACKWARD-UPDATE$(output, prompts_label)$
end function

function TEXT-PPL-LOSS$(output, prompts_label)$
    $loss \leftarrow CE(output, prompts_label)$
    $ppl \leftarrow e^{loss}$
    return ppl
end function

Table 4: Visualizing prompts trained by our PLPP

#	UCF101	Flowers102	Caltehc101	StanfordCars	FGVCAircraft
1	verify	hooper	learning	headquarters	humor
2	exhilarating	hike	6	minis	discuss
3	igne	beh	shoot	rebound	existed
4	discuss	accurate	reduces	ator	city
5	gi	core	uta	sport	pakistan
6	fx	signing	2	ville	direct
7	visited	vs	inspired	stats	imma
8	night	iq	fields	machine	ays
9	real	manager	cartoon	back	injury
10	drop	per	mark	amount	womensday
11	controls	hike	bubble	drew	ho
12	suggestions	website	hike	sa	hel
13	bec	year	babe	2	lemans
14	smile	und	threatened	herself	its
15	edge	2	shoutout	version	kick
16	jersey	seventeen	cs	colored	papa